

DEPARTMENT OF MINING ENGINEERING
Indian Institute of Technology (Indian School of Mines), Dhanbad
MID-SEMESTER EXAMINATION
Monsoon Semester

Examination: B-Tech (Mining).

Session: 2023-24 (16.09.2023).

Subject: Surface Mining (MNC – 300).

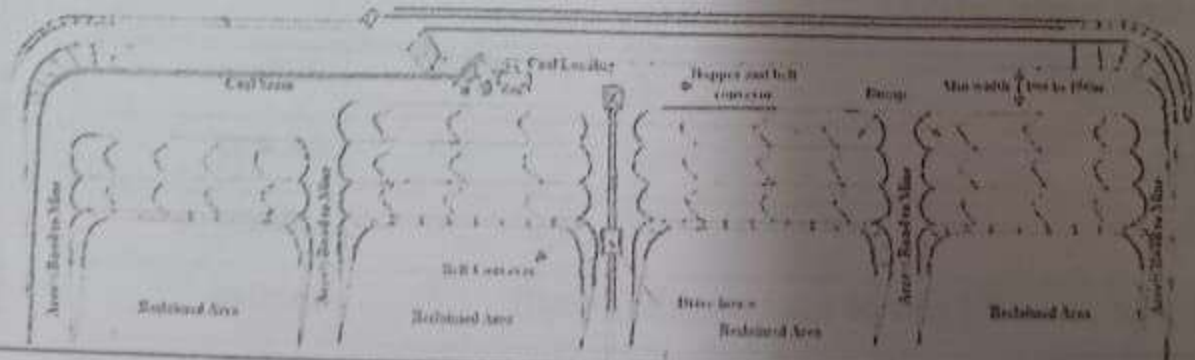
Duration: 2 Hours.

Instructions:

Make suitable assumptions if required.

Max. Marks: 60

Section A: Answer Any Two Questions

Sl. No.	Question	Marks
1	Explain the various unit operations of Surface Mining. Explain with the help of diagram unit operations of shovel-dumper system in a semicontinuous mode of unit operation.	8 + 7
2	What is a Push Back operation in Surface Mining? Explain, with the help pf diagrams, types of push back operations, their advantages, and disadvantages.	5 + 10
3	A typical coal mine with a strike length of 2 km is planned to produce 9.0 million tons of coal per annum in a semi-continuous mode of unit operations (Figure). The coal thickness is about 10 meters, and the specific gravity of coal is 1.5.  Three 100 tons in-pit trucks are used to feed the conveyor. Cost of one dumper is INR 30 million with an operating cost of INR 10000 per hour.	5 + 10

Initial cost of installation of conveyors system (idlers, motor, pulleys, shafts, crushers, feeders etc.) to start the operation is INR 500 million, and thereafter. the incremental cost of laying the conveyor system is @ Rs. 10 million for every 100m. Operating costs for belt conveyor system is as under.

- Repair and Maintenance – Rs. 50,000 per hour
- Power cost – Rs. 5000 per hour
- Man-power cost – Rs. 5000 per hour.
- Other support – Rs. 5000 per hour.

Assume

1. Annual operating hours for plants and machineries is 5000 hours,
2. Life of plants and machineries is 5 years.
3. Ignore depreciation values of plants and machineries.

Calculate.

1. Annual rate of advance of the mine along dip direction
2. Net Present Cost (NPC) of coal transport system at 15% rate of discount for the initial 5 years.

Section A: Answer Any Three Questions

Sl. No.	Question	Marks																		
1	A surface coal mine producing 10.0 Mt coal annually. Estimate the drill productivity and number of drills required for coal production at the following operating parameters. Specific gravity of coal as 1.5. Operating parameters: <table border="1"> <tr> <td>Bench Height</td><td>8</td><td>meters</td></tr> <tr> <td>Burden</td><td>5</td><td>meters</td></tr> <tr> <td>Spacing</td><td>6</td><td>meters</td></tr> <tr> <td>Length of one drill rod or feed length (l-feed)</td><td>5</td><td>meters</td></tr> <tr> <td>Time to change and un-change 1 change of drill rod (t-feed)</td><td>1</td><td>minute</td></tr> <tr> <td>Set up time for drill (T-set up)</td><td>3</td><td>minutes</td></tr> </table>	Bench Height	8	meters	Burden	5	meters	Spacing	6	meters	Length of one drill rod or feed length (l-feed)	5	meters	Time to change and un-change 1 change of drill rod (t-feed)	1	minute	Set up time for drill (T-set up)	3	minutes	5 + 5
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in	Average travel time of drill from one hole to another (T-travel)	2	minutes	
	Drilling rate (r-drill)	0.4	metre/minute	
	Annual availability (%)	80	%	
	Utilization	70	%	
	Operator's efficiency (annual basis)	80	%	
2	Briefly describe the major environmental impacts of surface mining.			10
3	i) Briefly explain, with examples, the key considerations for designing of dumping operationn in Surafec Mines? ii) Briefly explain the dumping methods used in surface mining operation.			7 + 3
4	Briefly describe the In-pit Crushing and Conveying (IPCC) systems and its advantages.			10